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Appraising energy, transport, and climate investments

Project appraisal, investment programming, socioeconomic development planning, policy review, inter-agency advisory work, and data analytics for the Philippine Government, 2015 to 2026

34
infrastructure projects, USD 14.05 billion

15
technical assistance and program financings

48
bills, laws, and policies reviewed

8
national boards and committees advised



Bernie Calderon Magtalas
Infrastructure economist and licensed electrical engineer
bcmagtalas.github.io/Career-Portfolio

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Published October 2026. All figures in United States dollars (USD) unless stated. Amounts for technical assistance and programs are each program's total approved amount. Select any chapter or annex above to go to it; links in the text open the cited sources. The interactive version of this report is at bcmagtalas.github.io/Career-Portfolio.

CHAPTER 1 · PROFILE AND CAREER PATH

From project analysis to investment and policy decisions

Bernie Calderon Magtalas is an infrastructure economist and licensed electrical engineer. He is a Supervising Economic Development Specialist in the Infrastructure Staff of the Department of Economy, Planning, and Development (DEPDev), where he leads project appraisal, planning, and policy formulation across the energy, transportation, and wider public infrastructure sectors.

He began his career at the Department of Energy (DOE) in December 2015, supporting renewable energy and rural electrification programs, and joined the National Economic and Development Authority (NEDA, now DEPDev) in September 2016.

CAREER PATH



Dec 2015 2018 2020 2022 2024 Present

- **Supervising Economic Development Specialist**
  DEPDev · January 2025 to present
- **Senior Economic Development Specialist**
  NEDA (now DEPDev) · January 2020 to December 2024
- **Economic Development Specialist**
  NEDA (now DEPDev) · September 2016 to December 2019
- **Science Research Specialist**
  Department of Energy · December 2015 to August 2016

CORE EXPERTISE

- Project economic analysis
- Cost-benefit analysis
- Infrastructure appraisal
- Investment programming
- PPP risk assessment
- R · Python · Excel
- Energy and transport policy
- Socioeconomic development planning

THE RECORD IN NUMBERS

34	projects appraised, USD 14.05 billion
15	programs reviewed, USD 1.77 billion
12	consulting studies, USD 12.68 million
48	bills, laws, and policies reviewed
8 + 7	boards advised and working groups served
207	projects prioritized in a USD 176 billion pipeline

CHAPTER 2 · APPRAISAL PORTFOLIO

USD 14.05 billion across 34 infrastructure projects

Appraisal covered financial and economic models; market, technical, social, environmental, and institutional analyses; gender responsiveness; and, for public-private partnerships, concession agreements, risk allocation, and prequalification.



PUBLIC-PRIVATE PARTNERSHIP
10,436.09
USD million · 74% · 16 projects

LOCAL FUNDS
2,698.14
USD million · 19% · 13 projects

OFFICIAL DEVELOPMENT ASSISTANCE
913.10
USD million · 7% · 5 projects

Value by sub-sector

SUB-SECTOR	USD M	PROJECTS
Airport	7,362.59	10
Roads and bridges	1,595.90	1
Rail	1,455.33	1
Energy	1,170.74	15
Urban	734.73	2
Ports	709.39	3
Expressways	546.96	1
Safety	471.69	1
Total	**14,047.33**	**34**

HIGHLIGHT · ENERGY

Agus-Pulangi hydropower rehabilitation, USD 512 million

He leads the PPP appraisal of the hydropower complex, a major source of clean power for the Mindanao grid.

HIGHLIGHT · TRANSPORT

PPP risk assessment, USD 10 billion in contracts

He led the risk assessment of eight PPP transport projects and built mitigation into their contracts.

Largest projects appraised 8 of 34 · 78% of total value

PROJECT	SUB-SECTOR	FINANCING	USD M
Mactan-Cebu International Airport integrated development plan	Airport	PPP	4,382.42
Quezon-Bicol Expressway	Roads and bridges	Local funds	1,595.90
Cebu Monorail Transit System	Rail	PPP	1,455.33
Laguindingan Airport	Airport	PPP	982.92
Davao International Airport	Airport	PPP	825.81
Metro Manila Intelligent Transport System	Urban	Local funds	610.07
Cavite-Tagaytay-Batangas Expressway	Expressways	PPP	546.96
Agus-Pulangi Hydroelectric Power Plants Rehabilitation	Energy	PPP	512.00

Full list of 34 projects with descriptions: Annex A

CHAPTER 3 · TECHNICAL ASSISTANCE AND PROGRAMS

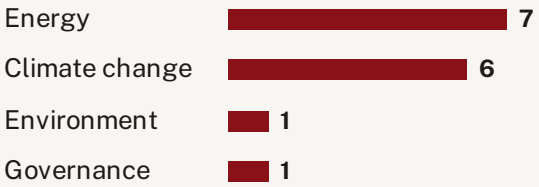
USD 1,774.48 million across 15 programs reviewed

He reviews technical assistance, grants, and policy-based loans in response to development partners' requests for no-objection, comments, and endorsement. Amounts are each program's total approved amount, including other participating countries for regional programs.

PROGRAMS AND POLICY-BASED LOANS
1,730.00
USD million · 3 programs

TECHNICAL ASSISTANCE AND GRANTS
44.48
USD million · 12 programs

Programs by sector



Programs and policy-based loans

Philippines Second Energy Transition and Climate Resilience Development Policy Loan World Bank (IBRD) loan and Livable Planet Fund grant · June 2026	1,020.00
Climate Change Action Program, Subprogram 3 ADB policy-based loan, cofinanced by AIIB, AFD, and JICA · October 2026 (DBCC)	650.00
Philippines Energy Threshold Program Grant from the US Millennium Challenge Corporation · June 2026	60.00

CONSULTING STUDIES

USD 12.68 million across 12 studies

He evaluated the technical and financial proposals for all 12 studies, from procurement through contract negotiation. For two studies, he also administered the consulting contract and evaluated the consultants' outputs.

MASTER PLANS
6.16
USD million · 6 studies

DEVELOPMENT PLANS
4.10
USD million · 3 studies

FEASIBILITY STUDIES
2.42
USD million · 3 studies

CONTRACT ADMINISTERED
Hybridization of existing diesel power plants with renewable energy and battery storage
Feasibility study · Region VIII · USD 0.75 million

CONTRACT ADMINISTERED
Sustainable urban infrastructure development master plan
Master plan · Vigan City · USD 0.81 million

Full lists with descriptions: Annexes B and C

CHAPTER 4 · POLICY AND GOVERNANCE

48 bills, laws, and policies reviewed; 8 boards advised

He has reviewed these measures for NEDA (now DEPDev) since 2016, and as a supervisor in the Infrastructure Staff since 2025, through position papers, comments, and inter-agency technical working groups.

32	11	21
energy measures	transportation measures	reviewed in 2025–2026 alone

Selected measures, 2025–2026

Philippine National Nuclear Energy Safety Act — LAW
Enacted as Republic Act No. 12305 (September 2025); creates the Philippine Atomic Energy Regulatory Authority

Amendments to the Electric Power Industry Reform Act — BILL
Strengthening the regulatory powers and capacity of the Energy Regulatory Commission

Philippine Nuclear Liability Act — BILL
House Bill No. 375, 20th Congress

Waste-to-Energy Regulatory Framework Act — BILL
19th and 20th Congress versions

Accelerating the Transition to Renewable Energy through Electric Vehicles in Mass Transport — POLICY
Draft administrative order

Carbon Pricing Instruments for the Philippines — POLICY
Policy review on climate change

BOARDS AND COMMITTEES ADVISED · 8

ENERGY POLICY AND SECURITY

Energy Investment Coordinating Council	EICC
Nuclear Energy Program Inter-Agency Committee	NEP-IAC
Inter-Agency Energy Efficiency and Conservation Committee	IAEECC
Inter-Agency Energy Contingency Committee	IECC
IECC Oil Contingency Task Force	IECC-OCTF

POWER SECTOR AND INFRASTRUCTURE

National Power Corporation Board	NPC
Power Sector Assets and Liabilities Management Corporation Board	PSALM
Infrastructure Development Committee	InfraCom

WORKING GROUPS · 7

SECTOR POLICY

Electric Vehicle Industry Development Act technical working group	EVIDA-TWG
Technical Panel for Evaluation of Innovative Energy Projects	
Sub-Committee on the SDGs, working group on environment	SC-SDGs
Philippines' COVID-19 Recovery Cluster	

WITHIN DEPDEV

Climate Change Community of Practice	COP
Bids and Awards Committee technical working group	BAC-GIP
Gender and Development Focal Point System	GAD

CHAPTER 5 · ANALYTICAL TOOLS

Built in R, Python, and Excel

FIRST IN THE PHILIPPINES

Regional Infrastructure Development Index

He directed five ADB consultants and five DEPDev specialists in computing the index in R. Its dashboard was adopted as a prototype.

MASTER'S THESIS

Structural Equation Modeling of Global Energy Transition

The site presents his factor analysis and structural equation models in R across 27 variables and 120 countries.

207 PROJECTS

Multi-criteria evaluation model

This automated Excel model prioritized projects in a USD 176 billion national investment pipeline.

KNOWLEDGE NETWORKS AND FORUMS



Panelist · 26 February 2026 · Bangkok

SIPA session at the 13th Asia-Pacific Forum on Sustainable Development

Sustainable Infrastructure, Resilient Links: Energy, Transport and Industry in Asia's Low-Carbon Future



Resource person and participant · 2025–2026

SIPA Southeast Asia Training Seminars, Asian Institute of Technology

Three in-person seminars in Thailand and nine webinars on infrastructure appraisal

Future Energy Leader (FEL-100) · 2023–2024

World Energy Council

Nominated with the support of the Japan Energy Association and selected at the 26th World Energy Congress, Rotterdam, April 2024; contributor to the World Energy Trilemma 2024

Panel discussant · 2024

Ease of Doing Business Forum 2024

Executive Order No. 59: streamlining the permitting process for infrastructure flagship projects

Panel discussant

Socioeconomic Development Planning Advocacy

Briefing with ADB and the University of Tokyo on infrastructure principles, policies, and processes

Co-Deputy Head for Education and Research

2030 Youth Force in the Philippines

Youth organization working toward the Sustainable Development Goals at the grassroots



FIELD MISSIONS, 2016

He took part in sustainability assessments of off-grid solar home system programs with DOE, the National Electrification Administration, and the World Bank, ahead of the scaled-up Access to Sustainable Energy Programme.

Negros Island Region · 8–12 July 2016 · Photovoltaic Mainstreaming projects, three electric cooperatives

Infanta, Pangasinan · 19–21 July 2016 · HEART and SOUL Project, 267 solar home systems

CHAPTER 6 · METHODS AND TOOLS

How projects and policies were assessed

These methods are applied under the NEDA (now DEPDev) Reference Manual on Project Development and Evaluation and the Investment Coordination Committee (ICC) guidelines.

APPRAISAL METHODS

Appraisal framework

Integrated project analysis across market, technical, financial, economic, institutional, social, and risk modules; incremental with-and-without-project comparison; least-cost alternatives across technology, location, scale, and timing

Economic analysis

Shadow pricing with ICC parameters (10% social discount rate, shadow exchange and wage rates); conversion factors, removal of transfers, valuation of externalities; EIRR, ENPV, and benefit-cost ratio

Financial analysis

Real and nominal cash-flow models from the owner, total-investment, and government budget viewpoints; FIRR against the weighted average cost of capital; annual liquidity and debt-service checks

Risk and sensitivity

ICC sensitivity cases (costs +10% and +20%, benefits −10% and −20%, combined) and switching values; PPP risk identification, allocation, and mitigation costing

Social and distributional

Stakeholder analysis reconciling financial and economic results; social impacts and willingness to pay

Climate and environment

Greenhouse gas abatement costing, climate resilience, and nature-related impacts

10%	+10, +20%	−10, −20%	7
social discount rate (ICC)	cost sensitivity cases	benefit sensitivity cases	modules in the appraisal framework

PLANNING, POLICY, AND RESEARCH

Investment programming
Multi-criteria analysis to prioritize projects in the national infrastructure investment pipeline

Policy analysis
Comparative analysis of legislation and regulatory frameworks, such as benchmarking the natural gas industry bill against the electric power industry's market framework

Statistics
Confirmatory factor analysis and structural equation modeling

TOOLS AND LANGUAGES

Analysis
R, Python, and Excel (economic and financial models)

Visualization
Interactive web dashboards built from R and Python outputs

Energy modeling
OSeMOSYS and FlexTool (Climate Compatible Growth training, 2025)

Languages and location
Filipino (native), English (fluent) · Pasig City, Philippines

CHAPTER 7 · EDUCATION AND CREDENTIALS

2024

Master of Arts in International Relations

Waseda University, Graduate School of Asia-Pacific Studies · Tokyo, Japan

September 2022 to September 2024 · Specialization in Economic Development and Policy in Asia

SCHOLARSHIP — Fully funded JICA Sustainable Development Goals Global Leader Program

THESIS — **Charting the Path to Net Zero:** confirmatory factor analysis and structural equation models in R across 27 variables and 120 countries on the drivers of the global energy transition

COURSEWORK — Analysis of Quantitative Data I and II · Multivariate Statistics · Data Science for Management · Environmental Economics and Environmental Policy · Sustainable Development and International Development Cooperation

2014

Bachelor of Science in Electrical Engineering

University of the Philippines Los Baños, College of Engineering and Agro-Industrial Technology

May 2008 to August 2014 · Major in Power Engineering

LICENSE AND CERTIFICATIONS

LICENSE
Registered Electrical Engineer
Professional Regulation Commission · April 2015

CERTIFICATION
Certified PPP Professional (CP3P), Foundation
APMG International · January 2021

CERTIFICATION
Certified Flexible Power System Specialist
Renewables Academy (RENAC) · November 2018

CERTIFICATION
Certified Green Finance Specialist
Renewables Academy (RENAC) · November 2017

COURSE
Energy and Flexibility Modelling: OSeMOSYS and FlexTool
Climate Compatible Growth · June 2025

COURSE
Modelling, Policy and Political Economy
Climate Compatible Growth · May 2025

Training continues on the next page · Certificates and academic records: Annex H

CHAPTER 7 · TRAINING

Sustainable Infrastructure Programme in Asia

SIPA is a regional knowledge network on sustainable infrastructure run by the Asian Institute of Technology, the United Nations Development Programme, and the OECD, funded by Germany's International Climate Initiative.

SOUTHEAST ASIA TRAINING SEMINARS

1 28 APRIL–2 MAY 2025
First SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, Asian Institute of Technology, Thailand · Certificate in Annex H

2 28 JULY–1 AUGUST 2025
Second SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, Asian Institute of Technology, Thailand · Certificate in Annex H

3 22–26 FEBRUARY 2026
Third SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, Asian Institute of Technology, Thailand · Certificate in Annex H

WEBINAR SERIES · 9 ONLINE SESSIONS, JUNE 2025 TO MARCH 2026

1	Financial Evaluation of Infrastructure Projects	24 June 2025
2	Economic Evaluation of Infrastructure Projects: Foundations for Integrating Social and Environmental Considerations	8 July 2025
3	Economic Evaluation of Infrastructure Projects: Approaches, Tools and Methods	22 July 2025
4	Planning Under Risk and Uncertainty: Scenario-Based Simulation for Infrastructure Viability	29 October 2025
5	Capturing the Full Value in Infrastructure Evaluation: Integrating Ecosystem Services, Social Well-Being, and Equity	19 November 2025
6	Pricing Infrastructure for Efficiency, Equity, and Sustainability	10 December 2025
7	Measuring the Ripple Effects of Infrastructure Development	16 January 2026
8	Institutions Matter: Strengthening Energy Governance and Policy to Close Infrastructure Finance Gap	18 February 2026
9	Integrated Framework for Sustainable Infrastructure Development	31 March 2026

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PORTFOLIO bcmagtalas.github.io/Career-Portfolio

Infrastructure projects appraised

34 projects with a total cost of USD 14,047.33 million, appraised for Investment Coordination Committee review. Costs in USD million.

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
1	**Access to Sustainable Energy Programme (ASEP): Investment Support Component** The overall objective of the ASEP is to assist the Government of the Philippines (GOP) in (a) expanding its sustainable energy generation to meet its economy's growing needs and (b) providing energy access to the poor and marginalized sectors. It specifically aims to assist the GOP in achieving the following: a. Meet rural electrification targets through renewable energy; b. Promote energy efficiency; c. Expand sustainable energy generation to meet the growing needs of the economy; and d. Provide energy access for the poor through affordable, disaster-resilient energy systems. The investment support component of the ASEP involves the supply, design, and installation of 7 × 1.0 MWp solar photovoltaic (PV) plants or Rural Network Solar (RNS) to various electric cooperatives in the Philippines.	Energy	ODA from the European Union	7.70
2	**Agus-Pulangi Hydropower Plant Complex (APHC) Rehabilitation under Series of Project (SOP) 1** The project aims to increase the service life and improve the operational efficiency and reliability of hydropower plants (HPPs) under SOP 1 (i.e., Agus IV, V, VI, and VII). The project has the following specific objectives: a. Restore rated capacities of HPPs; b. Upgrade HPPs to improve their operational performance and conditions by using state-of-the-art monitoring, control, protection, and dispatch systems; c. Improve the safety of the HPPs; d. Improve power generation output to meet the anticipated power demand resulting from the Mindanao-Visayas Interconnection Project (MVIP); and e. Increase the country's overall output of clean energy.	Energy	ODA from the World Bank	195.56
3	**Hybridization of Existing Diesel Power Plant in Zumarraga Island Mini-Grid with Solar PV, Wind Turbines, and Batteries** The project aims to hybridize the existing diesel power plant operating in the Zumarraga Island mini-grid with solar PV, wind turbines, and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	7.74
4	**Hybridization of Existing Diesel Power Plant in Tagapul-an Island Mini-Grid with Biomass and Batteries** The project aims to hybridize the existing diesel power plant operating in the Tagapul-an Island mini-grid with biomass and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	1.12

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
5	**Hybridization of Existing Diesel Power Plant in Santo Niño Island Mini-Grid with Biomass, Solar PV, and Batteries** The project aims to hybridize the existing diesel power plant operating in the Santo Niño Island mini-grid with biomass, solar PV, and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	0.90
6	**Hybridization of Existing Diesel Power Plant in San Vicente Island Mini-Grid with Biomass, Solar PV, and Batteries** The project aims to hybridize the existing diesel power plant operating in the San Vicente Island mini-grid with biomass, solar PV, and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	0.81
7	**Hybridization of Existing Diesel Power Plant in San Antonio Island Mini-Grid with Biomass, Solar PV, and Batteries** The project aims to hybridize the existing diesel power plant operating in the San Antonio Island mini-grid with biomass, solar PV, and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	1.79
8	**Hybridization of Existing Diesel Power Plant in Capul Island Mini-Grid with Biomass and Batteries** The project aims to hybridize the existing diesel power plant operating in the Capul Island mini-grid with biomass and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	1.34
9	**Hybridization of Existing Diesel Power Plant in Camandag Island Mini-Grid with Biomass and Batteries** The project aims to hybridize the existing diesel power plant operating in the Camandag Island mini-grid with biomass and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	0.87
10	**Hybridization of Existing Diesel Power Plant in Biri Island Mini-Grid with Solar PV, Biomass, Wind Turbines, and Batteries** The project aims to hybridize the existing diesel power plant operating in the Biri Island mini-grid with solar PV, biomass, wind turbines, and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	4.51
11	**Hybridization of Existing Diesel Power Plant in Batag Island Mini-Grid with Biomass and Batteries** The project aims to hybridize the existing diesel power plant operating in the Batag Island mini-grid with biomass and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	0.73
12	**Hybridization of Existing Diesel Power Plant in Almagro Island Mini-Grid with Biomass and Batteries** The project aims to hybridize the existing diesel power plant operating in the Almagro Island mini-grid with biomass and batteries to improve the level of service and reduce the electricity cost in the area.	Energy	Local funds	0.67

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
13	**Unsolicited Operate-Add-Transfer (OAT) Proposal for the Laguindingan Airport** The unsolicited project involves the following components: a. Take-over of the operation and maintenance (O&M) of the existing facilities of Laguindingan Airport, i.e., from the Civil Aviation Authority of the Philippines (CAAP); b. Execution of the upgrade works, which include initial terminal expansion and refurbishment of the existing terminal; c. Future capacity augmentations through expansion or construction of a new passenger terminal and other buildings, along with all associated infrastructure and facilities per applicable standards; d. Installation of all required equipment and associated facilities per applicable standards; e. Enhancement or development of airside facilities to meet the enhanced scale of operations at the airport; and f. O&M of all enhanced airside facilities, including the apron, runway, and taxiway.	Airport	PPP (Unsolicited)	982.92
14	**Unsolicited Operate-Add-Transfer (OAT) Proposal: New Bohol International Airport** The proposed project involves the following components: a. Take-over of the O&M of the existing facilities of New Bohol International Airport from the Department of Transportation (DOTr); b. Capacity augmentation through expansion or construction of new passenger terminals, along with all associated infrastructure and facilities as per applicable standards once the existing terminal reaches its capacity (i.e., individual peak hour flows for particular facilities and processes are at least 70 percent of the design capacity); c. Installation of all the required equipment and associated facilities as per applicable standards; d. Enhancement or development of airside facilities to meet the enhanced scale of operations at the airport; and e. O&M of all enhanced airside facilities, including the apron, runway, and taxiway.	Airport	PPP (Unsolicited)	361.05
15	**Unsolicited Proposal for the Puerto Princesa International Airport** The project includes the following: a. Renovation, development, and expansion of the existing passenger terminal building (PTB) of the Puerto Princesa International Airport (PPIA) and its associated facilities per international civil aviation standards to accommodate the demand; b. Construction of an apron adjacent to the new PTB extension; c. Renovation, development, or expansion of commercial assets, cargo handling facilities, and discretionary facilities; d. Planning, design, and expansion of all utility systems necessary for undertaking the works and construction of the commercial assets; e. Operation and maintenance of the concessionaire O&M facilities and the commercial assets; and f. Procurement, maintenance, and installation of all information technology and other equipment required for the proper O&M of the airport, excluding air navigation systems, air traffic control systems, and security systems and equipment.	Airport	PPP (Unsolicited)	107.61

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
16	**Unsolicited Operate-Add-Transfer (OAT) Proposal for the Iloilo International Airport** The project includes the following: a. Expansion of capacity of the PTB of the Iloilo International Airport (IIA); b. Construction of an apron adjacent to the new PTB extension; c. Renovation/maintenance of the existing terminal building; d. Installation of the required equipment for the passenger services in the new PTB; and e. O&M of the passenger terminal buildings and equipment/facilities in the PTBs.	Airport	PPP (Unsolicited)	126.14
17	**Unsolicited Proposal for the Development, Operation and Management of the Bacolod-Silay International Airport** The proposed project involves the development, operation, and management of the Bacolod-Silay International Airport and all necessary development works, such as construction, rehabilitation, improvement, expansion, modernization, and decommissioning needed to meet the projected demand requirement for the airport services within the 35-year concession period.	Airport	PPP (Unsolicited)	213.21
18	**Unsolicited Operate-Add-Transfer (OAT) Proposal for the Davao International Airport** The project involves developing, operating, and managing the Davao (F. Bangoy) International Airport. The scope includes undertaking the operation, management, and maintenance of all existing and project assets within the airport's boundary, except those remaining within CAAP's control. It also includes all necessary development works, such as construction, rehabilitation, improvement, expansion, and modernization, needed to meet the projected demand for airport services within the concession period.	Airport	PPP (Unsolicited)	825.81
19	**Unsolicited Operate-Maintain-Upgrade-and-Transfer (OMUT) Proposal for the Kalibo International Airport** The proposed project involves the upgrade, O&M, and future expansions of the Kalibo International Airport. It specifically includes the following components: a. Take-over of the O&M of the current airport facility; and b. Implementing a facility upgrade of the airport, including systems and construction of the PTB.	Airport	PPP (Unsolicited)	63.84
20	**Unsolicited Proposal for the Cavite-Tagaytay-Batangas Expressway (CTBEx) Project** The project is a 50.42-kilometer (km), four-lane toll road linking the Cavite-Laguna Expressway (CALAx) at Silang East Interchange to Tagaytay City and Nasugbu, Batangas. It aims to provide the following: a. An alternative route to the traffic plying Aguinaldo Highway and Santa Rosa–Tagaytay Road; b. Linkage between Metro Manila, Cavite, and Batangas; and c. High-standard highway with increased accessibility to nearby areas such as Amadeo and Indang.	Expressways	PPP (Unsolicited)	546.96

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
21	**Unsolicited Proposal for the Port of General Santos** The project involves the redevelopment and O&M of the Port of General Santos. It includes port development and modernization, delivery of reefer container services, and use of modern information technology systems.	Ports	PPP (Unsolicited)	101.54
22	**Unsolicited Proposal for the Iloilo Commercial Port Complex (ICPC) and Port of Dumangas (POD)** The project involves the operation and management of the container terminal of ICPC and PTB and the roll-on-roll-off (RORO) operations of POD. It also involves the provision of the following: a. Wharf extension of 250 meters to support international ship operations; b. Modern fleet of equipment; c. Power and reefer facilities; d. Storage management; and e. Information technology network and infrastructure, terminal operating system, and ancillary systems.	Ports	PPP (Unsolicited)	225.62
23	**Unsolicited Proposal for the Port of Davao (Sasa)** The project involves the following: a. Rehabilitation and reconfiguration of the port's infrastructure facilities to accommodate modern equipment and systems and increase the overall capacity of the port to serve Panamax vessels and over 1,300,000 twenty-foot equivalent units (TEUs) of container traffic and general cargoes; b. Installation of modern port equipment that will significantly enhance the level of service of the port, thereby reducing dwell time and contributing to lower transportation costs; c. Introduction of port systems to ensure sustainability while enhancing efficiency; d. Use of innovative technologies, including smart gate system and radio-frequency identification (RFID) truck driver tags, and terminal appointment and booking system; and e. Professionalized O&M of the port for a period of 25 years.	Ports	PPP (Unsolicited)	382.23
24	**Unsolicited Build-Transfer-Operate and Maintain (BTOM) Proposal: Cebu Monorail Transit System Project** The proposal involves the implementation of a 27-km straddle-type monorail system for the Metro Cebu area composed of the following segments: a. A 17.7 km monorail transit system (i.e., Cebu Central Line) consisting of an elevated structure and a section with 14 stations, including a short tunnel; b. A 9.3-km monorail transit system (i.e., Cebu Airport Line) consisting of an elevated structure and an improved bridge crossing between Mactan Island and the mainland consisting of 6 stations; and c. A 5-hectare (ha) depot facility located close to the Mactan-Cebu International Airport Station.	Rail	PPP (Unsolicited)	1,455.33
25	**Quezon-Bicol Expressway Project** The project will start at Pagbilao, Quezon, and will end at the existing Maharlika Highway in San Fernando, Camarines Sur. The project will have an indicative length of approximately 180 km and will be an alternative option for travel from Quezon and the Bicol provinces.	Roads and Bridges	Local funds	1,595.90

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
26	**New Dumaguete (Bacong) Airport** The proposed New Dumaguete Airport will cover 197.55 ha of land development to be located in the Municipality of Bacong, Negros Oriental in the Negros Island Region (NIR). The scope of works will cover detailed engineering design (DED), civil works, construction supervision services, project management, land acquisition and compensation, and maintenance supervision.	Airport	ODA from KEXIM	274.84
27	**Unsolicited Build-Operate-Transfer (BOT) Proposal for the 50-Year Integrated Development Plan for the Mactan-Cebu International Airport** The proposed project involves the phased implementation of the following activities: a. Airfield upgrade works; b. Airfield capacity augmentation; c. New passenger terminal building (PTB) and related capacity augmentation; and d. Other capital expenditures and medium- to long-term enhancements of the airfield.	Airport	PPP (Unsolicited)	4,382.42
28	**Unsolicited Proposal for the Busuanga Airport** The project involves the O&M of the landside facilities of the existing Busuanga Airport. The scope also includes undertaking necessary improvements of the airport facilities and services to ensure compliance with minimum performance standards and specifications.	Airport	PPP (Unsolicited)	24.75
29	**Acquisition of Forty (40) Units of Fast Patrol Craft (FPC)** The proposed project involves the acquisition of the following: a. Forty units of 30- to 35-meter (m), weapon-ready Multi-Purpose Fast Response Boats (MPFRBs) or Fast Patrol Craft (FPC) with ballistic protection, i.e., 20 units to be built in the Philippines and 20 units to be built abroad; b. Nine years of Integrated Logistics Support (ILS), i.e., spare parts and maintenance services; and c. Support equipment for six Philippine Coast Guard (PCG) bases.	Safety	Local funds	471.69
30	**Metro Manila Intelligent Transport System (MMITS)** The project involves establishing the Metro Manila ITS as a unified management system to ease congestion in Metro Manila by utilizing different strategies through ITS subsystems.	Urban	Local funds	610.07
31	**Unsolicited Build-Own-Operate (BOO) Proposal for the North Luzon Express Terminal (NLET) Project** The proposed project is an integrated transport terminal facility for northbound provincial bus routes designed to be a multi-level, mixed-use terminal facility, hosting mainly provincial buses of northern destinations. Aside from terminal operations, the site would also host office spaces, rental and commercial areas serving integrated transport terminal users, transfer facilities for feeder modes, and parking facilities.	Urban	PPP (Unsolicited)	124.66

NO.	PROJECT AND DESCRIPTION	SUB-SECTOR	FINANCING	USD M
32	**Philippine Geothermal Resource Derisking Facility (PGRDF)** The facility aims to reduce the financial risks of early-stage geothermal exploration and attract private investment in greenfield geothermal development. The Department of Energy (DOE) will implement the facility through the Land Bank of the Philippines (LANDBANK), covering up to 50 percent of eligible exploration drilling costs through sub-loans that developers repay only if exploration confirms a commercially viable resource. The appraisal covered economic, financial, social, and risk analyses, project alternatives, demand assumptions, greenhouse gas abatement costs, and nature-related impacts, and secured Investment Coordination Committee (ICC) approval.	Energy	ODA from the ADB	175.00
33	**Agus-Pulangi Hydroelectric Power Plants Rehabilitation and Asset Management Project** The project aims to rehabilitate the hydroelectric power plants of the Agus-Pulangi Hydropower Complex in Mindanao and establish an asset management framework that restores their rated capacities, extends their service life, and improves the reliability of clean power supply in the Mindanao grid. The appraisal covered economic, financial, social, and risk analyses, project alternatives, demand assumptions, greenhouse gas abatement costs, and nature-related impacts. The appraisal is ongoing.	Energy	PPP	512.00
34	**National Energy Efficiency and Conservation Program (NEECP)** The program aims to accelerate the adoption of energy efficiency and conservation measures in government buildings and facilities, reducing public-sector energy consumption and costs. The appraisal covered economic, financial, social, and risk analyses, project alternatives, and greenhouse gas abatement costs; the program awaits ICC approval.	Energy	ODA from the ADB	260.00
	Total			**14,047.33**

Technical assistance, grants, and programs reviewed

Reviewed in response to development partners' requests for no-objection, comments, and endorsement. Amounts are each program's total approved amount; most technical assistance projects are regional, so the amount covers all participating countries.

B.1 Technical assistance and grants (12)

NO.	PROGRAM	SECTOR	APPROVED	USD M
1	**Advance Sustainable Clean Energy Network for Development (ASCEND)** ADB TA 10312-REG · ADB Technical Assistance Special Fund (TASF) and Regional Cooperation and Integration Fund, cofinanced by the Korea e-Asia and Knowledge Partnership Fund and the Japan Fund for Prosperous and Resilient Asia and the Pacific Regional TA covering 11 countries, with the Philippines added through supplementary financing. It helps build diversified and responsible supply chains from critical minerals to clean energy manufacturing through knowledge products, policy dialogue, capacity building, and a pipeline of investment projects.	Energy	Jun 2024	4.65
2	**Development of Renewable Energy Public-Private Partnership Projects and Transaction Advisory Services** ADB TA 10375-REG · ADB Technical Assistance Special Fund (TASF) Regional transaction advisory TA that helps scope, prepare, structure, and tender renewable energy, energy storage, and transmission public-private partnership (PPP) projects, with a Philippine renewable energy project among the candidate transactions.	Energy	Nov 2024	3.00
3	**Accelerating Climate Action, Innovation, and Private Sector Development through Sector Operations** ADB TA 10406-REG · ADB TASF Regional TA covering 39 member countries, including the Philippines. It funds diagnostics, research, and capacity building to grow pipelines of climate action, private sector development, and advanced technology projects, and to support countries' Paris Agreement commitments.	Climate Change	Nov 2024	14.80
4	**Advancing Energy Transition and Regional Power Trade in the Greater Mekong Subregion and Southeast Asia** ADB TA 10480-REG · ADB TASF, cofinanced by the ASEAN Catalytic Green Finance Facility trust funds, the High Level Technology Fund, and the Clean Energy Fund Regional TA covering eight countries, including the Philippines. It supports the energy transition and multilateral power trade under the ASEAN Power Grid through regional market frameworks, country action plans, and the design of an ASEAN Power Grid financing facility.	Energy	Dec 2024	5.00
5	**Adaptation Metrics to Scale Up and Align Investment with Climate-Resilient Development** ADB TA 10048-REG · ADB TASF TA covering Bangladesh, Pakistan, and the Philippines. It helps governments and the private sector use adaptation metrics in planning and link them to public and private finance. The Philippine consultation mission and national co-design workshop were held in August 2025.	Climate Change	Dec 2022	1.00

NO.	PROGRAM	SECTOR	APPROVED	USD M
6	**Scaling Up Clean Air Actions for Better Health and Resilience** ADB TA 10537-REG · ADB TASF, cofinanced by the Japan Fund for the Joint Crediting Mechanism and the Urban Resilience Trust Fund Regional TA covering 12 countries, including the Philippines, under ADB's Asia Clean Blue Skies Program. It supports clean air policy, planning, and capacity, and helps design and finance clean air investments.	Environment	Apr 2025	4.13
7	**Operations, Knowledge, and Analytics Support in Fragile and Conflict-Affected Situations and Small Island Developing States** ADB TA 10045-REG · ADB TASF Regional TA, extended to the Philippines, that builds capacity to apply differentiated approaches in fragile and conflict-affected situations, including fragility and resilience assessments, analytical tools, and knowledge sharing.	Governance	Dec 2022	3.75
8	**Promoting Upstream Low-Carbon and Sustainable Environment Solutions in Southeast Asia** ADB TA 10633-REG · ADB TASF, cofinanced by the UK-ASEAN Catalytic Green Finance Facility Regional TA covering seven Southeast Asian countries, including the Philippines. It builds a pipeline of climate-friendly projects through guidance on green ports and new green and transition finance models for governments and financial institutions.	Climate Change	Nov 2025	1.50
9	**Catalyzing Growth and Emerging Areas in Asia and the Pacific** ADB TA 0078-REG · Smart Energy Innovation Fund Regional TA cluster covering Azerbaijan, Mongolia, the Philippines, and Thailand. It supports integrated and smart energy solutions, including smart grids, energy storage, and electric mobility, through diagnostics, project preparation, and knowledge transfer.	Energy	Nov 2025	4.00
10	**Readiness Support to Strengthen the Philippines' Capacity and Knowledge on Accessing the Green Climate Fund** GCF PHL-RS-003 · Green Climate Fund Readiness grant, delivered by the Global Green Growth Institute (GGGI) Builds government capacity to access Green Climate Fund financing through training, knowledge products, a country portfolio analysis, and climate finance tracking systems. DEPDev commented on the draft Country Portfolio Analysis Report in 2025.	Climate Change	Sep 2019	0.30
11	**Strengthening the Capacity of Institutions in the Philippines to Comply with the Transparency Requirements of the Paris Agreement (Capacity-building Initiative for Transparency, CBIT Philippines)** GEF ID 11950 · Global Environment Facility grant through Conservation International; USD 2.15 million cofinancing Strengthens institutions and greenhouse gas measurement and reporting systems so the Philippines can meet the Paris Agreement's enhanced transparency requirements. Executed by the Climate Change Commission and the Manila Observatory.	Climate Change	Jan 2026	1.95

NO.	PROGRAM	SECTOR	APPROVED	USD M
12	**Advancing Renewable Energy through Floating Solar Photovoltaics in the Philippines** GGGI PH047 · Korea Green New Deal Trust Fund, delivered by GGGI Prepares the first bankable utility-scale floating solar project on a National Irrigation Administration dam and lays the policy and regulatory groundwork to replicate it across more than 100 reservoirs.	Energy	Started Jun 2025	0.40
	Total			**44.48**

B.2 Programs and policy-based loans (3)

NO.	PROGRAM	SECTOR	APPROVED	USD M
1	**Climate Change Action Program (CCAP), Subprogram 3** ADB policy-based loan · Policy-based loan from the ADB, cofinanced by the Asian Infrastructure Investment Bank (AIIB), Agence Française de Développement (AFD), and the Japan International Cooperation Agency (JICA) The policy-based loan supports national reforms to mainstream climate change adaptation and mitigation in public planning, budgeting, and investment. The assessment of the program informed the Government of the Philippines in its loan negotiations with the cofinancing institutions.	Climate Change	Oct 2026 (DBCC)	650.00
2	**Philippines Second Energy Transition and Climate Resilience Development Policy Loan** World Bank P510184 · International Bank for Reconstruction and Development (IBRD) loan of USD 1 billion and a USD 20 million Livable Planet Fund grant Supports reforms to scale up clean energy, including the renewable energy market and the first offshore wind auction, make electricity markets more secure and competitive, and improve water governance and tariffs.	Energy	Jun 2026	1,020.00
3	**Philippines Energy Threshold Program** Millennium Challenge Corporation (MCC) · Grant from the US Millennium Challenge Corporation Addresses high and unreliable power costs by helping electric cooperatives, streamlining Energy Regulatory Commission processes, and supporting smart grid and private energy investment.	Energy	Jun 2026	60.00
	Total			**1,730.00**

Consulting studies

Evaluated the technical and financial proposals for all 12 studies, from procurement through contract negotiation. For two studies (Nos. 2 and 5), also administered the consulting contract and evaluated the consultants' outputs. Costs in USD million.

NO.	STUDY	LOCATION	USD M
1	**Feasibility Study (F/S) for the Pasig River Ferry System Project** The study aims to determine the viability of the Pasig River Ferry System Project to improve intra-city connectivity within Metro Manila and regional connectivity among Rizal, Cavite, and Bulacan. It also aims to mitigate traffic congestion and air pollution in the regions covering these provinces.	National Capital Region (NCR)	0.58
2	**F/S on Hybridization of Existing Diesel Power Plants (DPPs) with Renewable Energy (RE) and Battery** Contract administered The study aims to evaluate the potential for hybridization of existing DPPs with RE (e.g., solar, wind, hydro) and battery in the island grids and missionary areas in Region VIII or Eastern Visayas. The study also aims to lay the groundwork for implementing the identified viable hybrid power plant projects, especially by electric cooperatives (ECs) willing to take over the off-grid generation function in missionary areas covered by their respective franchises.	Region VIII	0.75
3	**Master Plan for the Sustainable Urban Infrastructure Development in Metropolitan Cagayan de Oro** The studies aim to formulate master plans to guide decision-makers in preparing and approving plans, policies, and programs, activities, and projects (PAPs). Business cases or pre-feasibility studies (pre-F/S) will be formulated for the top five infrastructure PAPs that will be identified, and a GIS-based map will be generated to overlay these projects with existing facilities/developments in the study areas.	Region X	1.06
4	**Master Plan for Sustainable Urban Infrastructure Development in General Santos City** One of the sustainable urban infrastructure master plans (see No. 3).	General Santos City	1.12
5	**Master Plan for the Sustainable Urban Infrastructure Development in Vigan City** Contract administered One of the sustainable urban infrastructure master plans (see No. 3).	Vigan City	0.81
6	**Master Plan for the Sustainable Urban Infrastructure Development in Butuan City** One of the sustainable urban infrastructure master plans (see No. 3).	Butuan City	1.06
7	**Master Plan for the Sustainable Urban Infrastructure Development in Metropolitan Tuguegarao** One of the sustainable urban infrastructure master plans (see No. 3).	Metropolitan Tuguegarao	1.08
8	**National Broadband Plan (NBP) – Accelerated Fiber Build (Access) Project** The project will provide a comprehensive and detailed F/S for establishing and constructing fiber optic cables or a hybrid of other technologies deemed appropriate and associated facilities (e.g., conduits). The F/S will cover the appraisal of the project's technical, economic, and financial viability, identification of the government's actual needs, which favor socioeconomic over commercial services (e.g., GDP growth, education, e-health, surveillance and security, registration, and tax), identification of implementation variants/models and risks, and preparation of preliminary engineering design, among others.	Nationwide	1.09

NO.	STUDY	LOCATION	USD M
9	**Tourism Infrastructure Development Plans in Top Tourist Destinations in the Philippines Based on the Carrying Capacities** The objective of the study is to determine the carrying capacities of the top tourist destinations in the Philippines and formulate business cases or pre-F/S for the top three types of tourism-supporting infrastructure (e.g., water supply, sanitation, and sewerage; solid waste management; traffic management; and connectivity) for each tourist destination. The study will also generate GIS-based maps to overlay these projects with existing facilities/developments.	Nationwide	1.56
10	**Tourism Infrastructure Development Plans for Transforming Communities Towards Resilient, Inclusive and Sustainable Tourism (TouRIST) Program** The project aims to identify and prioritize the essential infrastructure for resilient, inclusive, and sustainable tourism within the framework of updated or duly formulated Local Tourism Infrastructure Development Plans for Siargao Island, Siquijor, Coron and El Nido, Palawan, Davao City and Samal Island, and Bohol. It will have a companion pre-F/S of top-priority PAPs on the identified sites.	Siargao Island, Siquijor, Coron and El Nido, Palawan, Davao City and Samal Island, and Bohol	1.48
11	**Integrated Infrastructure and Investment Corridors in Central Luzon** The study aims to formulate an integrated infrastructure and investment development plan (3IDP) that would promote the growth of investments in Central Luzon and identify and develop the infrastructure needed to support this growth.	Region III	1.06
12	**Master Plan for Sustainable Infrastructure Development in Olongapo and its Immediate Environs** The study aims to formulate a master plan for Olongapo and its neighboring areas to guide decision-makers in preparing and approving PAPs. Business cases or pre-F/S will be formulated for the top three infrastructure PAPs that will be identified, and GIS-based maps will be generated to overlay these projects with existing facilities/developments.	Olongapo and its immediate environs	1.03
	Total		**12.68**

Bills, laws, and policies reviewed

48 measures reviewed for NEDA (now DEPDev) since 2016, and as supervisor in the Infrastructure Staff from 2025, through position papers, comments, and inter-agency technical working groups.

D.1 Measures reviewed in 2025–2026 (21)

NO.	MEASURE	DESCRIPTION	SECTOR
1	**Amendments to the Electric Power Industry Reform Act (EPIRA): Strengthening the Energy Regulatory Commission** Bill · Senate and House measures, 20th Congress	Amends the Electric Power Industry Reform Act of 2001 (Republic Act No. 9136) to strengthen the regulatory powers and institutional capacity of the Energy Regulatory Commission.	Energy
2	**Waste-to-Energy Regulatory Framework Act** Bill · 19th and 20th Congress versions	Sets a national energy policy and regulatory framework for facilities that use waste-to-energy technologies, implemented with local governments. A LEDAC priority measure.	Energy
3	**Philippine National Nuclear Energy Safety Act** Law · Senate Bill No. 2899 / House Bill No. 9293, 19th Congress; enacted as Republic Act No. 12305 (18 September 2025)	Provides a comprehensive legal framework for nuclear safety, security, and safeguards in the peaceful use of nuclear energy, and creates the Philippine Atomic Energy Regulatory Authority (PhilATOM).	Energy
4	**Expansion of the Franchise Area of Davao Light and Power Company, Inc.** Bill · House Bill No. 11072, 19th Congress	Expands the franchise area of Davao Light and Power Company, Inc. to additional areas in the Davao Region.	Energy
5	**Philippine Nuclear Liability Act** Bill · House Bill No. 375, 20th Congress	Sets the rules on civil liability and compensation for nuclear damage: who is liable, and how affected parties can claim compensation for incidents at nuclear installations or during the transport of nuclear materials.	Energy
6	**Incentives for the Integration of the Nuclear Energy Sector into the Electricity Sector** Bill · House Bill No. 8567, 20th Congress	Prescribes incentives for integrating nuclear energy into the country's electricity sector and appropriates funds for this purpose.	Energy
7	**Amendments to the Biofuels Act of 2006 (Republic Act No. 9367)** Bill · Pending House Bills, 20th Congress	Amends the Biofuels Act of 2006 (Republic Act No. 9367), which mandates the blending of biofuels into locally sold gasoline and diesel.	Energy
8	**Amendments to the Downstream Oil Industry Deregulation Act (Republic Act No. 8479) and a Strategic Fuel Reserve** Bill · Senate and House measures, 20th Congress	Amends the Downstream Oil Industry Deregulation Act of 1998 (Republic Act No. 8479) and establishes a strategic fuel reserve to manage supply and price disruptions.	Energy
9	**Sariling Kuryente Act: Behind-the-Meter Generation, Energy Storage, and Rooftop Solar** Bill · Substitute bill, House Committee on Energy, 20th Congress	Promotes self-generation of electricity by households and businesses through behind-the-meter generation, energy storage systems, and rooftop solar.	Energy

NO.	MEASURE	DESCRIPTION	SECTOR
10	**Franchise Renewal of Visayan Electric Company, Inc.** Bill · Senate Committee on Public Services, 20th Congress	Renews the legislative franchise of Visayan Electric Company, Inc. to distribute electricity in its Metro Cebu service area.	Energy
11	**Franchise Coverage Expansion of MORE Electric and Power Corporation** Bill · Senate Committee on Public Services, 20th Congress	Expands the franchise coverage of MORE Electric and Power Corporation, the electricity distributor of Iloilo City.	Energy
12	**Draft Administrative Order: Accelerating the Transition to Renewable Energy through the Use of Electric Vehicles in Mass Transport** Policy · Office of the President; Department of Transportation	Accelerates the shift to renewable energy by promoting the use of electric vehicles in public mass transport.	Transportation
13	**IAEECC Resolution No. 13: Guidelines for Government Entities as Eligible End-Users under the Customer Choice Programs in the Retail Market** Policy · Inter-Agency Energy Efficiency and Conservation Committee; signed September 2026	Sets guidelines allowing government entities to choose their electricity supplier as eligible end-users under the customer choice programs of the retail electricity market.	Energy
14	**IAEECC Resolution Directing Government Entities to Prioritize the Purchase or Rental of Electric Vehicles** Policy · Inter-Agency Energy Efficiency and Conservation Committee	Directs government entities to prioritize electric vehicles when buying or renting motor vehicles, in line with the Electric Vehicle Industry Development Act (Republic Act No. 11697).	Energy
15	**Amendments to the Electric Vehicle Recognition Guidelines (Department Circular No. DC2023-05-0012)** Policy · Department of Energy	Amends the Department of Energy's guidelines for recognizing and classifying electric vehicles under the Electric Vehicle Industry Development Act.	Energy
16	**Framework for the Integration of Nuclear Energy in the Clean Energy Scenario of the Philippine Energy Plan 2023–2050** Policy · Department of Energy, draft Department Circular	Sets the framework for integrating nuclear power into the clean energy scenario of the Philippine Energy Plan 2023–2050.	Energy
17	**Nuclear Capacity Tender Guidelines** Policy · Department of Energy, draft Department Circular	Sets the guidelines for competitive tenders to procure nuclear generating capacity.	Energy
18	**National Nuclear Security Policy and Strategy** Policy · NEP-IAC Subcommittee 2	Sets the national policy and strategy for securing nuclear and radioactive materials and facilities.	Energy
19	**Amendments to the Implementing Guidelines of the Philippine Transport Vehicles Fuel Economy Labeling Program** Policy · Department of Energy	Revises the registration procedures and compliance mechanisms of the program that requires fuel economy labels on vehicles.	Transportation

NO.	MEASURE	DESCRIPTION	SECTOR
20	**Amendment to DHSUD Department Circular No. 2024-004: Locational Clearance for Projects of National Significance** Policy · Department of Human Settlements and Urban Development	Amends the guidelines for processing locational clearances for projects declared of national significance.	Governance
21	**Carbon Pricing Instruments for the Philippines** Policy · Department of Finance; Carbon Pricing Support TWG	Assesses legal and policy options for carbon pricing instruments in the Philippines.	Climate Change

D.2 Earlier measures (27)

NO.	MEASURE	TITLE	SECTOR
22	**Waste-to-Energy Act** Senate Bill No. 363	An Act Establishing a National Energy Policy and Regulatory Framework for Facilities Utilizing Waste-to-Energy Technologies	Energy
23	**Waste-to-Energy Act of 2019** Senate Bill No. 401	An Act Allowing the Use of Waste to Energy Technology in Electricity, Fuel and Heat Generation, and for other Purposes	Energy
24	**Rural Airport Development Act** House Bill No. 5075	An Act Granting the Right to Franchisees to Develop the Rural Airport Infrastructure Network in the Philippines for Commercial and for Other Purposes	Transportation
25	**Organic Law for the Bangsamoro Autonomous Region in Muslim Mindanao** Republic Act No. 11054	An Act Providing for the Organic Law for the Bangsamoro Autonomous Region in Muslim Mindanao, Repealing for the Purpose Republic Act No. 6734, Entitled “An Act Providing for an Organic Act for the Autonomous Region in Muslim Mindanao,” as Amended by Republic Act No. 9054, Entitled “An Act to Strengthen and Expand the Organic Act for the Autonomous Region in Muslim Mindanao”	Governance
26	**One Million Solar Roofs Act of 2017** House Bill No. 6565	An Act Providing for Incentives and Credit Facilities for Consumers of Small Solar Power Systems in the Philippines	Energy
27	**Senate Bill No. 1007**	An Act to Reconstruct and Rehabilitate the Existing 479-kilometer PNR lines from San Fernando, La Union to Legazpi City, Albay, Providing Funds from the Annual Value-Added Tax Collections, Amending for the Purpose the National Internal Revenue Code of 1997, As Amended	Transportation
28	**Philippine Downstream Natural Gas Industry Development Act (enacted as RA No. 12120, the Philippine Natural Gas Industry Development Act, 8 January 2025)** House Bill No. 10649	An Act Promoting the Development of the Philippine Downstream Natural Gas Industry, Consolidating for the Purpose All Laws Relating to the Transmission, Distribution and Supply of Natural Gas, and Appropriating Funds Therefor	Energy

NO.	MEASURE	TITLE	SECTOR
29	**Senate Resolution No. 359**	Resolution Directing the Senate Committee on Energy to Conduct an Inquiry in Aid of Legislation on the Sufficiency of the Department of Energy's Existing Policy on the Minimum Inventory Requirement for Oil and Petroleum Products and the Necessity of Establishing a Strategic Petroleum Reserve to Ensure the Security of the Country's Oil Supply	Energy
30	**Senate Resolution No. 360**	A Resolution Directing the Senate Committee on Energy to Conduct an Inquiry on the Implementation of Republic Act No. 8479 Otherwise Known as the Downstream Oil Industry Deregulation Act of 1998 With the End in View of Introducing Remedial Legislation as May Be Necessary	Energy
31	**Air Carriers Relief Act** House Bill No. 9324	An Act Providing for Relief Package to the Air Transport Sector of the Philippines	Transportation
32	**Electric Vehicles and Charging Stations Act** Senate Bill No. 1382	An Act Providing the National Energy Policy and Regulatory Framework for the Use of Electric Vehicles and the Establishment of Electric Charging Stations	Energy
33	**Electric Vehicle Industry Development Act** House Bill No. 10213	An Act Providing for the Development of the Electric Vehicle Industry and Establishing a Regulatory Framework for the Manufacture and Use of Electric Vehicles	Energy
34	**Energy Efficiency and Conservation Act** Republic Act No. 11285	An Act Institutionalizing Energy Efficiency and Conservation, Enhancing the Efficient Use of Energy, and Granting Incentives to Energy Efficiency and Conservation Projects	Energy
35	**Philippine Energy Research and Policy Development Institute Act of 2016** Senate Bill No. 1290	An Act Establishing the Philippine Energy Research and Policy Development Institute, Defining Its Objectives, Powers and Functions, Providing Funds Therefor, and for Other Purposes	Energy
36	**Tax Reform for Acceleration and Inclusion (TRAIN)** Senate Bill No. 1592	An Act Amending Sections 5, 6, 24, 25, 31, 34, 35, 51, 79, 84, 86, 89, 90, 97, 99, 100, 101, 106, 107, 108, 109, 112, 114, 116, 148, 149, 150, 151, 155, 171, 196, 232, 237, 249, 264, and 288; Creating New Sections 148-A, 150-A, 237-A, 264-A, 264-B, and 265-A; All Under Republic Act No. 8424, Otherwise Known as the National Internal Revenue Code of 1997, as Amended, and for Other Purposes	Trade and Industry
37	**The Solar Energy in National Government Offices Act of 2016** House Bill No. 2172	An Act Reducing the Cost of Electricity for National Government Offices by Promoting the Development and Utilization of Solar Energy in All Government Building and Offices, Providing Funds Therefor, and for Other Purposes	Energy
38	**Solar Energy Act of 2016** House Bill No. 5402	An Act Mandating Government Offices and/or Establishments to Allot at Least Thirty Percent (30%) of their Energy Requirements to Solar Energy, and Corollarily, Mandating the Same to Install Solar Panels unto Their Buildings	Energy

NO.	MEASURE	TITLE	SECTOR
39	**Small Solar Power System Incentives Act** House Bill No. 6960	An Act Promoting the Use of Renewable Energy in Homes by Providing Incentives and Credit Facilities for Consumers Utilizing Small Solar Power Systems	Energy
40	**House Bill No. 8723**	An Act Strengthening the Protection and Management of Waterways and Water Systems by Host Communities, and for Other Purposes	Water
41	**House Bill No. 1484**	An Act Requiring All Independent Power Producers, Generation Companies or Energy Resource Developers to Remit the Amount They Are Required to Set Aside as Financial Benefit Directly to the Host Community	Energy
42	**House Bill No. 1388**	An Act Providing for the Development, Rehabilitation and Modernization of Naval Airport in the Province of Biliran and to Allocate Funds Thereof	Transportation
43	**Ubay Airport Development Act** House Bill No. 2112	An Act Providing for the Ubay Airport Development to Accommodate Commercial Domestic and Aviation Training Flights, Appropriating Funds Therefor, and for Other Purposes	Transportation
44	**House Bill No. 2769**	An Act Providing for the Establishment and Construction of an International Airport in the Center of Mindanao, Particularly in the Are Between Damulog, Bukidnon in the North; Antipas, North Cotabato in the East; Carmen, North Cotabato in the South; And Alamada, North Cotabato in the West, and Appropriating Funds Therefor	Transportation
45	**House Bill No. 4238**	An Act Providing for the Establishment and Construction of an Airport in the Municipality of Luuk, Province of Sulu, for the Benefit of the Entire Province of Sulu, to Be Known as Luuk Airport, and Appropriating Funds Therefor	Transportation
46	**House Bill No. 4393**	An Act Designating Mati Airport in Davao Oriental as the Official Alternate Airport to the Davao International Airport and Appropriating Funds Therefor	Transportation
47	**House Bill No. 7847**	An Act Establishing a Domestic, Commercial Airport in the Municipality of San Francisco, Province of Cebu, and Appropriating Funds Therefor	Transportation
48	**The Maria Cristina Falls Act of the Philippines** House Bill No. 2317	An Act Guaranteeing the Right to the Natural Wealth and Granting a Twenty Percent Discount (20%) in the Electricity Rates to All Electric Consumers in Iligan City, Lanao Del Norte, And Lanao Del Sur being the Host City and Provinces of Maria Cristina Falls, Agus River, and Lake Lanao	Energy

Boards, committees, and working groups

National boards and committees advised, and working groups served, with each body's mandate.

E.1 Boards and committees advised (8)

NO.	BODY, LEGAL BASIS, AND MANDATE
1	**Energy Investment Coordinating Council (EICC)** Energy policy and security · Legal basis: EO 30, s. 2017 The EICC was created through Executive Order (EO) No. 30, s. 2017, entitled "Creating the Energy Investment Coordinating Council in Order to Streamline the Regulatory Procedures Affecting Energy Projects." It is mandated to spearhead and coordinate national government efforts to harmonize, integrate, and streamline regulatory processes, requirements, and forms relevant to the development of energy investments in the country, primarily with regard to Energy Projects of National Significance (EPNS), to facilitate transparency and accountability among concerned agencies. EPNS are major energy projects for power generation, transmission, and/or ancillary services, including those required to maintain grid stability and security.
2	**Nuclear Energy Program Inter-Agency Committee (NEP-IAC)** Energy policy and security · Legal basis: EO 116, s. 2020 The NEP-IAC was constituted through Executive Order (EO) No. 116, entitled "Directing a Study for the Adoption of a National Position on a Nuclear Energy Program, Constituting a Nuclear Energy Program Inter-Agency Committee, and for Other Purposes." It is mandated to spearhead the study for the adoption of a national position for an NEP. Under EO No. 164, s. 2022, which adopted the national position on the nuclear energy program, the NEP-IAC was further directed to perform the functions of a Nuclear Energy Program Implementing Organization (NEPIO) as recommended by the International Atomic Energy Agency (IAEA).
3	**Inter-Agency Energy Efficiency and Conservation Committee (IAEECC)** Energy policy and security · Legal basis: RA 11285 Republic Act (RA) No. 11285, or the Energy Efficiency and Conservation (EEC) Act, created the IAEECC. It aims to assess and approve the government's energy efficiency projects and deliver strategic direction in executing the Government Energy Management Program (GEMP). GEMP is a government-wide program that aims to reduce the government's monthly electricity and petroleum product consumption through electricity and fuel efficiency/conservation.
4	**Inter-Agency Energy Contingency Committee (IECC)** Energy policy and security · Legal basis: AO 6, s. 2011 The IECC was organized through Administrative Order (AO) No. 6, series of 2011, entitled "Organizing the Inter-Agency Energy Contingency Committee (IECC)." This is to evaluate and examine the government's energy strategies addressing contingencies arising from the energy supply disruptions.
5	**Inter-Agency Energy Contingency Committee-Oil Contingency Task Force (IECC-OCTF)** Energy policy and security · Legal basis: IECC Resolution 1 The IECC-OCTF was created through IECC Resolution No. 1, entitled "Creation of the Oil Contingency Task Force," to ensure an adequate and continuous supply of high-quality petroleum products amidst the threat of oil disruption in the country.
6	**National Power Corporation (NPC) Board of Directors** Power sector and infrastructure · Legal basis: RA 9136 (EPIRA) The NPC Board of Directors was reconstituted through RA No. 9136, otherwise known as the Electric Power Industry Reform Act (EPIRA) in 2001, to govern the following functions of the National Power Corporation (NPC): (i) missionary electrification function through the Small Power Utilities Group (SPUG); (ii) management of the watersheds; and, (iii) the efficient and effective management of the government's remaining power assets.
7	**Power Sector Assets and Liabilities Management Corporation (PSALM) Board** Power sector and infrastructure · Legal basis: RA 9136 (EPIRA) The PSALM Board was established through RA 9136 to govern the orderly sale and privatization of all existing generation assets of the NPC, independent power producer (IPP) contracts, real estate, and all other disposable assets, including the transmission business of the National Transmission Corporation (TransCo). It also governs the liquidation of all NPC's financial obligations.

NO.	BODY, LEGAL BASIS, AND MANDATE
8	**Infrastructure Development Committee (InfraCom)** Power sector and infrastructure · Legal basis: RA 12145 The Infrastructure Development Committee (InfraCom) is a committee of the Economy and Development (ED) Council. It was originally established as the NEDA Committee on Infrastructure under Executive Order (EO) No. 230, s. 1987, and was reconstituted under Section 29 of Republic Act (RA) No. 12145, the Economy, Planning, and Development Act (2025), which repealed EO 230 and reorganized the National Economic and Development Authority into DEPDev. It is co-chaired by the Secretaries of Public Works and Highways and of Economy, Planning, and Development, with the Executive Secretary and the Secretaries of Agriculture, Budget and Management, Environment and Natural Resources, Human Settlements and Urban Development, Information and Communications Technology, Interior and Local Government, Energy, Finance, Tourism, Transportation, and Trade and Industry as members, as further reorganized by Administrative Order No. 37, s. 2025. It advises the President and the ED Council on matters concerning infrastructure development; coordinates the activities of agencies, including government-owned or controlled corporations concerned with infrastructure development; and recommends to the President government policies, programs, and projects concerning infrastructure development consistent with national development objectives and priorities.

E.2 Working groups (7)

NO.	BODY AND MANDATE
1	**Electric Vehicle Industry Development Act (EVIDA)-TWG** Sector policy · Legal basis: RA 11697 The EVIDA-TWG was created through Republic Act (RA) No. 11697 or the Electric Vehicle Industry Development Act (EVIDA) to ensure harmonized policies and consistent and streamlined standards and regulations in the EV industry.
2	**Climate Change Community of Practice (COP)** Within DEPDev · Legal basis: NEDA SO 160-2022 The COP was established through National Economic and Development Authority Special Order 160-2022 — the Authority having since been reorganized into the Department of Economy, Planning, and Development (DEPDev) — to serve as a platform for DEPDev staff and offices to exchange, develop, and communicate knowledge and best practices on climate change. The COP is also responsible for mainstreaming climate change into DEPDev's internal processes.
3	**Technical Panel for Evaluation of Innovative Energy Projects** Sector policy The technical panel evaluates energy projects that exhibit novelty in implementation/service delivery or solutions that improve or replace existing products, services, or processes. These include, among others, wave energy facilities, solar-powered bulk water systems, integrated solar-powered water/irrigation pumps, solar-powered weather stations, and advanced device and materials testing laboratories for battery technologies.
4	**Gender and Development (GAD) Focal Point System** Within DEPDev The GAD focal point system prepares DEPDev's GAD plan and budget and ensures GAD mainstreaming in the government's plans, projects, programs, and activities.
5	**DEPDev Bids and Awards Committee for Goods and Infrastructure Projects (BAC-GIP) TWG** Within DEPDev The BAC-GIP evaluates the technical, financial, and legal bid proposals for the goods and infrastructure projects that DEPDev procures.
6	**Sub-Committee on Sustainable Development Goals (SC-SDGs) TWG on Environment** Sector policy The TWG refines and monitors the SDG indicators on the environment and reviews SDG-related budgets and programs.
7	**Philippines' COVID-19 Recovery Cluster** Sector policy The Philippines' COVID-19 Recovery Cluster plays a significant role in setting the direction of the country's economic recovery from COVID-19. It formulates and implements action plans and economic policies to restart the country's social and economic activities.

Engagements, training, and knowledge exchange

Knowledge networks, forums, and training. The Sustainable Infrastructure Programme in Asia (SIPA) is run by the Asian Institute of Technology, the United Nations Development Programme, and the OECD, funded by Germany's International Climate Initiative.

F.1 Forums, networks, and organizations

DATE	ENGAGEMENT AND ROLE
26 February 2026	**SIPA session at the 13th Asia-Pacific Forum on Sustainable Development, Bangkok** Panelist Sustainable Infrastructure, Resilient Links: Energy, Transport and Industry in Asia's Low-Carbon Future.
28 April–2 May 2025	**First SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development** Resource person and participant In person, Asian Institute of Technology, Thailand
28 July–1 August 2025	**Second SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development** Resource person and participant In person, Asian Institute of Technology, Thailand
22–26 February 2026	**Third SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development** Resource person and participant In person, Asian Institute of Technology, Thailand
2024	**Ease of Doing Business Forum 2024** Panel discussant Executive Order No. 59: Streamlining the Permitting Process for Infrastructure Flagship Projects
2023–2024	**World Energy Council** Future Energy Leader (FEL-100) Selected at the 26th World Energy Congress, Rotterdam, 22–25 April 2024; regional commentary contributor to the World Energy Trilemma 2024 report
—	**Socioeconomic Development Planning Advocacy** Panel discussant Briefing on infrastructure development principles, policies, and processes with the Asian Development Bank and the University of Tokyo
—	**2030 Youth Force in the Philippines Inc.** Co-Deputy Head for Education and Research Youth organization working toward the Sustainable Development Goals at the grassroots

F.2 SIPA webinar series (9 sessions, online)

NO.	SESSION	DATE
1	Financial Evaluation of Infrastructure Projects	24 June 2025
2	Economic Evaluation of Infrastructure Projects: Foundations for Integrating Social and Environmental Considerations	8 July 2025
3	Economic Evaluation of Infrastructure Projects: Approaches, Tools and Methods	22 July 2025
4	Planning Under Risk and Uncertainty: Scenario-Based Simulation for Infrastructure Viability	29 October 2025

NO.	SESSION	DATE
5	Capturing the Full Value in Infrastructure Evaluation: Integrating Ecosystem Services, Social Well-Being, and Equity	19 November 2025
6	Pricing Infrastructure for Efficiency, Equity, and Sustainability	10 December 2025
7	Measuring the Ripple Effects of Infrastructure Development	16 January 2026
8	Institutions Matter: Strengthening Energy Governance and Policy to Close Infrastructure Finance Gap	18 February 2026
9	Integrated Framework for Sustainable Infrastructure Development	31 March 2026

Field mission reports, 2016

Sustainability assessments of off-grid solar home system (SHS) programs, prepared while at the Department of Energy ahead of the scaled-up Access to Sustainable Energy Programme (ASEP).

G.1 · NEGROS ISLAND REGION

Sustainability Assessment of Photovoltaic Mainstreaming (PVM) Projects in Negros Island Region

DATES	LOCATION	TEAM	SITES
8–12 July 2016	Negros Island Region	DOE, NEA, and the World Bank	4 sites, 3 electric cooperatives

SITE 1 Aluyan, Caduhaan, Cadiz City

Northern Negros Electric Cooperative (NONECO)

Technical

- The supplier delivered and installed substandard wires.
- Batteries and CFLs are the most common components that fail.
- There are private suppliers of cheaper, substandard SHS.
- The beneficiaries are touching and tweaking the system.

Maintenance

- SHS beneficiaries are too far from the electric cooperative. Thus, they could not report their concerns regarding the failing SHS equipment to the EC.
- Only one technician was trained within the sitio.
- NONECO is not visiting the consumers.
- The supplier did not provide after-sales services to the beneficiaries.

Feedback on the new SHS package

- The visited households are receptive to the new package to be offered.
- Some are requesting SHS with a larger capacity to accommodate other appliances like refrigerators, TVs, and DVD players.



The access roads to the sitio proper are under construction. Owner-type jeepneys and motorcycles are the common modes of transportation.

Bill Collection

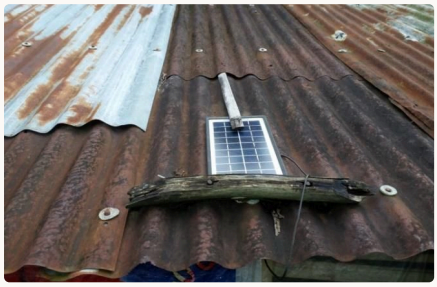
- The collected payments are being forwarded to NONECO through the barangay captain.
- The households are paying inconsistently. They are paying different amounts at random months of the year.
- The households only pay to a collector with a valid NONECO ID.

Ownership

- Some beneficiaries thought that the barangay captain owned the systems.
- Rumors are circulating that the government owns the SHSs. Thus, the systems should be free.




The solar panels installed in each household are in good condition. However, most of the batteries already failed. Some households have additional batteries connected to the system.





The connections of the system are very poor. One inspected household has a battery directly connected to the solar panel. The battery is prone to overcharging and exploding in this type of connection. Meanwhile, most households replaced the busted CFLs with any bulbs available. Some households bought substandard solar panels with smaller capacity.

SITE 2 Nailab, Ara-al, La Carlota City

Negros Occidental Electric Cooperative (NOCECO)

Technical

- The wires are oversized.
- The supplier delivered CFLs instead of LEDs.
- The SHSs have poor connections.
- The solar panels are facing in different directions.
- Some beneficiaries are knowledgeable about the technology.

Bill Collection

- The beneficiaries pay PHP 275 monthly.
- The beneficiaries stop paying when the bulbs fail. They want the EC to fix the busted SHS component before continuing the monthly payment.
- The payment of each beneficiary is not being monitored. The collector does not issue individual receipts.
- The collector forwards the total collected payments to the EC. The EC then issues a receipt.
- The collector has a ledger with a list of SHS users.

Maintenance

- The trained local technician has no tools for SHS maintenance.
- Lack of access roads affects the maintenance of SHS.

Ownership

- The beneficiaries know that NOCECO owns the systems.

Feedback on the new SHS package

- The beneficiaries are receptive to the new package.




Sitio Nailab is about a 42-minute walk from the last area with an access road. The team had to cross a river via an improvised bridge to reach the sitio.



The wires of the inspected SHSs are oversized. Upon checking the connections, the team's technical consultants observed that the wires were not properly secured. Some strands of wire cannot fit through the intended holes for connection.

SITE 3 Sibucao, Mambucsay, Cauayan

Negros Occidental Electric Cooperative (NOCECO)

Technical

- Most of the SHSs are not functional.
- Some of the controllers delivered in 2010 were defective.
- One of the households has two SHSs.

Bill Collection

- The collector does not have a salary.
- There were two collectors of monthly bills. The barangay captain collected the payments in 2010, while the treasurer of the SOPA collected in 2013.
- The collector issues individual receipts.

Maintenance

- The beneficiaries replace the batteries and bulbs on their own.
- There are trained technicians. However, the equipment supplier is far from the site.

Ownership

- The recipients of SHS thought that the systems belonged to the government.
- Some thought the barangay captain owned the system because he was collecting the monthly bill.

Feedback on the new SHS package

- One of the respondents does not want to avail of the new SHS package because of the poor performance of the SHS currently installed.



The household on the left has two SHSs. The respondent stated that the additional SHS belongs to her relative. Meanwhile, although the household owners in the center have a decent income, they do not want to use the new SHS package because of the poor performance of the SHS currently installed. There are clustered households within the sitio. Generally, the residents have fairly good incomes.

SITE 4 Sitios Ilaya, Libtacon, and Mantaliok

Negros Oriental I Electric Cooperative (NORECO I)

The results of the ocular inspection and interviews in Sitios Ilaya, Libtacon, and Mantaliok are the same. NORECO I institutionalized the SHS's operation, payment, and ownership structure.

Technical

- The condition of SHS is rather good.
- The beneficiaries are not tampering with the systems.
- There are private suppliers of cheaper, substandard SHS.

Bill Collection

- NORECO I assigned a staff member to collect the monthly bill.
- There is a logbook to record the monthly payments.

Maintenance

- NORECO I pulled out the SHSs of those who can no longer pay and installed them sporadically for those who want to use the system.
- The beneficiaries are willing to retain the system even when connected to the grid.
- The local technician was able to fix minor damage to SHSs.

Feedback on the new SHS package

- The beneficiaries are willing to accept the new package.

Ownership

- The beneficiaries know that the systems belong to NORECO I.



The household owner on the left bought additional solar panels with a smaller capacity to accommodate additional loads. Meanwhile, the SHS in the center is in very good condition, and the local technician replaced only the bulbs. Households already connected to the grid, like the household on the right, still have their SHS installed.

Conclusions

The following are the factors that affect the implementation of PVM:

- Interventions of political entities
- Unsystematic billing and collection system
- Distances between the households
- Noncompliance with after-sales agreement, quality assurance, specifications, and standards of SHS
- Distance of the supplier from the sitios
- Confusion regarding the ownership of SHS
- Lack of personnel who will monitor the project implementation
- Access road
- Information dissemination
- Affordability of the system

Recommendations

1. DOE should establish sound selection criteria for SHS beneficiaries.
2. Consider the technology proposed by the consultants, which will map the households in remote and unviable areas.
3. Arrange a sitio-level organization that will monitor the implementation of the program.
4. The recipients of SHSs should be in clusters. This will solve the difficulty of collecting monthly bills.
5. Suppliers who delivered substandard SHS equipment and those who did not provide after-sales services should not be allowed to participate in the bidding process of the scaled-up implementation of PVM.
6. The electric cooperative may store spare SHSs within the EC office to provide an accessible supply of components.
7. The ECs must institutionalize the recipients regarding PVM. Seminars must be conducted before the installation of the units. Regular meetings within the sitios should be organized to monitor and assess program implementation.
8. The ECs should establish a dedicated unit or office for PVM.
9. The sitio-level organization should be capacitated to implement the program.
10. Coordinate with other sectors to develop a program with a holistic approach.
11. If possible, integrate electrification programs with livelihood and infrastructure projects.

G.2 · INFANTA, PANGASINAN

Sustainability Assessment of the Household Electrification Assistance through Renewable Technology and Social Preparation for the Upliftment of Lives (HEART and SOUL) Project in Preparation for the Scaled-Up Implementation of the ASEP-PVM

DATES	LOCATION	TEAM	PROJECT
19–21 July 2016	Barangay Pita, Infanta, Pangasinan	DOE, NEA, the World Bank, and IED	267 solar home systems (TEFI)

Introduction

- Household Electrification Assistance through Renewable Technology and Social Preparation for the Upliftment of Lives (HEART and SOUL) Project is an initiative of Team Energy Foundation, Inc. (TEFI) in response to the government's call to achieve 90 percent household electrification by 2017.
- In August 2015, TEFI installed 267 30-watt-peak photovoltaic solar home systems (SHSs) in two barangays in Infanta, Pangasinan: 164 in Pita and 103 in Babuyan. The SHS package includes four five-watt light bulbs, a provision for cellphone charging, and a digital AM-FM radio. The foundation also installed two solar-powered streetlights in each barangay and SHSs in their barangay halls and health centers.
- On 19–21 July 2016, representatives from the Department of Energy (DOE), National Electrification Administration (NEA), World Bank, and Innovation Energie Développement (IED) assessed the HEART and SOUL Project in Barangay Pita in preparation for the scaled-up implementation of the ASEP-PVM.

Objectives

- To determine the factors affecting the sustainability of the HEART and SOUL Project
- To learn good practices in project implementation, which can be integrated with the scaled-up implementation of the ASEP-PVM

Methodology

- Representatives from TEFI briefed the barangay captain and some residents of Pita, Infanta, Pangasinan, regarding the HEART and SOUL Project assessment.
- The team conducted technical inspections and interviews on random households. The samples chosen were based on the physical structure of the house.
- The technical and logistic issues affecting the project's sustainability were determined based on the information given by the respondents and technical inspection of the SHS packages.

Results and Discussion

Technical

- The majority of the surveyed households have worn-out batteries because of improper connection. According to the local technician, most beneficiaries opened the system's protective box. They then connected their loads directly to the battery terminals to bypass the controller and use the battery repeatedly beyond its threshold level, reducing its lifespan. Meanwhile, those who can afford to replace the battery connected it directly to the solar panel terminals. In this type of connection, the batteries may explode when overcharged.
- It was also observed that the solar panels are facing in different directions. According to the respondents, they oriented their unit to the direction of the sun to increase its output.

Bill Collection

- Approximately ten percent of the SHS customers are currently paying. Some can no longer afford the PHP 200 monthly maintenance and replacement fee, while the majority refuse to pay because of worn-out batteries. Due to the decreasing number of paying customers, the collector stopped the house-to-house collection. Consequently, those who want to pay must go to the collector's house. In any case, the collector constantly issues receipts to the paying customers for monitoring.

Maintenance

- There are three SHS technicians in the area. They repair minor damage to the SHSs and report to the supplier the components that need replacement. However, a considerable quantity of SHS components must fail before the supplier delivers the replacements. It usually takes two to three months to completely restore the system, and only paying customers can avail of this service.

Management

- There was good information dissemination regarding the HEART and SOUL Project. TEFI conducted a seminar before the SHSs were installed to explain the billing method, regulations, and technical know-how about the system. The foundation also organized a barangay association to administer the maintenance and replacement of SHS components and collect the monthly fee. Customers can access the maintenance and replacement fund at one percent monthly interest.

Conclusion

- The beneficiaries have basic knowledge about the technology. However, they are unaware of the risks of altering the SHS. The team observed that improper connections are the most common cause of damage in the system components. Meanwhile, tilting the solar panels towards the sun, though it can increase the energy produced by SHS, exposes the solar cells to mechanical damage. Both practices significantly affect the SHS's efficiency and, consequently, its sustainability.
- The availability of SHS components also affects the sustainable implementation of the project, especially the collection of fees. The assessment shows that aside from affordability, the majority of customers refuse to pay because of damaged equipment. The collected fees from the remaining paying customers will not be sufficient to sustain the SHSs.
- The HEART and SOUL Project demonstrates good practices that can be integrated with scaled-up implementation of the PVM: TEFI institutionalized its beneficiaries, organized an association, conducted social preparation, and provided a small-scale lending system to its beneficiaries. These factors could be substantial in constructing the framework of the ASEP-PVM. However, the gaps that the team determined should also be addressed to guarantee its efficient implementation.

Recommendation

1. The association should submit a monthly report indicating SHS customers' financial, technical, and social concerns.
2. Regular assembly should be conducted as part of the program's institutionalization.
3. The government should implement a strict policy regarding altering SHS connections and educate the beneficiaries regarding its risks. On the first offense, the customers who modify the connections should be given a warning; on the second offense, their unit should be pulled out.
4. Spare SHS components should be provided and stored within the EC office to restore the damaged systems immediately.
5. Livelihood and other infrastructure projects may be integrated into the program.

License, certificates, and academic records

REPUBLIC OF THE PHILIPPINES
PROFESSIONAL REGULATION COMMISSION
MANILA

EXAM NO. 07459309
DATE OF RELEASE APR. 30, 2015

BOARD OF ELECTRICAL ENGINEERING

MAGTALAS, BERNIE CALDERON

SIR/MADAM:

THE RATINGS YOU OBTAINED IN THE ***REGISTERED ELECTRICAL ENGINEERS*** LICENSURE EXAMINATION GIVEN BY THE BOARD IN MANILA, BAGUIO, C.D.O., ON APRIL 26 & 27, 2015 ARE AS FOLLOWS:
CEBU, DAVAO, ILOILO, LEGAZPI & LUCENA

SUBJECTS	RELATIVE WT.	RATINGS
1. MATHEMATICS - Algebra, Trigonometry, Analytical Geometry, Differential Calculus, Integral Calculus, Differential Equations, Complex Numbers, Probability and Statistics, Advanced Engineering Mathematics including Matrices, Power Series, Fourier Analysis, Laplace Transforms and others.	25	90
2. ENGINEERING SCIENCES AND ALLIED SUBJECTS - General Chemistry, College Physics, Computer Fundamentals and Programming, Engineering Materials, Engineering Mechanics, Fluid Mechanics, Strength of Materials, Thermodynamics, Electrical Engineering Law, Engineering Economics, Engineering Management, Contracts and Specifications, Code of Professional Ethics, Philippine Electrical Code, Parts 1 and 2 and others.	30	80
3. ELECTRICAL ENGINEERING PROFESSIONAL SUBJECTS - Electric Circuits, Electronic Theory and Circuits, Energy Conversion, Power Transmission and Distribution, Instrumentation and Measurement, Circuit and Line Protection, Control Systems, Principles of Communication, Electrical Machines, Electrical Equipment, Components and Devices, Electric Systems, Power Plant, Electronic Power Equipment, Illumination, Building Wiring and others.	45	86
GENERAL WEIGHTED AVERAGE RATING	100	85.20
REMARKS		PASSED

IMPORTANT:

1. *THIS REPORT IS NOT VALID IF THERE IS ANY ALTERATION.*
2. To pass the examination, an examinee must obtain a GENERAL WEIGHTED AVERAGE RATING of at least 70% with no rating below 50% in any subject/s. An examinee who obtains a general weighted average rating of 70% or higher but gets a rating below 50% in any given subject/s shall be allowed to re-take any number of times, but only in the subject/s where he/she obtained below 50% . In order to pass the examination, an examinee must obtain a rating of at least 70% in the subject(s) repeated. Ratings enclosed in parenthesis indicate subjects passed in a previous examination.

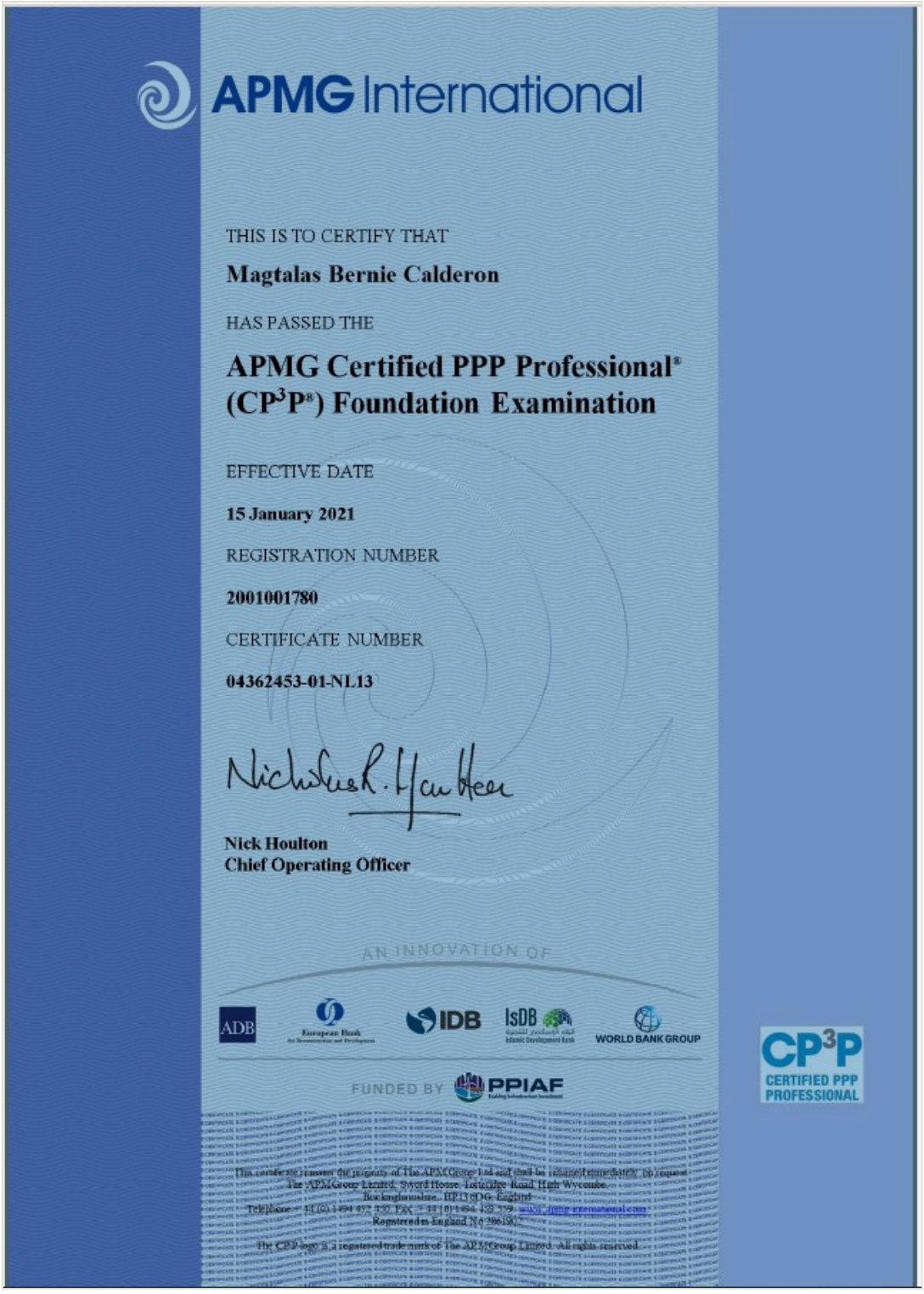
VERY TRULY YOURS,
FOR CHAIRMAN FLORENTINO C. DOBLE, M.D.

BY:

FRANCIS V. MAPILE
(CHAIRMAN, BOARD OF ELECTRICAL ENGINEERING)

15 - 0183912

H.1 Registered Electrical Engineer
Professional Regulation Commission, Philippines · April 2015



H.2 Certified PPP Professional (CP3P), Foundation
APMG International · January 2021



CERTIFICATE

renac
renewables academy

Certified Flexible

Power System Specialist (CFPSS)

Bernie Magtalas

successfully completed the above 180 hour online training

on 27 November 2018.

The final exam was passed with a mark of 94% (very good).

The online training covered the following topics:

- **Wind power** (physical basics, wind turbine components, wind farm planning and economic aspects)
- **Photovoltaic** (application, system configuration, energy yield estimation and economic aspects)
- **PV-diesel hybrid systems** (system components, sizing, dynamic system behavior, economic evaluation and case studies)
- **Wind and PV grid integration** (time series of variable renewable energy (wind and solar), system operation, balancing power, grid congestion management, capacity planning, grid code development, grid and system integration studies)
- **Flexibility options for power systems** (power system transformation, flexibility needs, flexibility options, levelised cost of flexibility and market framework)
- **Flexible thermal power plants** (flexibility criteria for new power plants, retrofit measures for existing power plants, market environment and case studies)

Berlin, 30 November 2018

Place, date

Berthold Breid, CEO

Supported by:

Federal Ministry for the Environment, Nature Conservation and Nuclear Safety

based on a decision of the German Bundestag

CapREG

Renewables Academy (RENAC) AG • Berlin, Germany • www.renac.de • ISO 9001 and ISO 29990 certified

To check the authenticity of this certificate, please contact verify@renac.de

H.3 Certified Flexible Power System Specialist (CFPSS)
Renewables Academy (RENAC) · November 2018



CERTIFICATE

renac
renewables academy

Green Finance Specialist

Blended learning programme on green energy and climate finance

Bernie Magtalas

successfully completed the above 200 hours online training and 3 days seminar in Manila, Philippines

from 26th June – 30th November 2017

and passed with a final mark of 82.7%.

The online training and seminar covered the following topics:

- Renewable energy and energy efficiency technology, market conditions and project specifics
- Support mechanisms for renewable energy and energy efficiency, market frameworks
- Financing of energy efficiency projects and systematic energy saving approaches
- Renewable energy project finance, debt financing process and project contracts
- Financial modelling of a renewable energy project using the "RE Project Evaluator" (MS Excel)
- Insurance as well as environmental and social aspects in renewable energy projects
- International, regional and domestic climate finance options
- Renewable energy projects in the portfolio context

Berlin, 30 November 2017
Place, date

Berthold Breid, CEO

green banking

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To check the authenticity of this certificate, please contact verify@renac.de

H.4 Certified Green Finance Specialist
Renewables Academy (RENAC) · November 2017



H.5 Energy and Flexibility Modelling: OSeMOSYS and FlexTool
Climate Compatible Growth · June 2025



H.6 Modelling, Policy and Political Economy
Climate Compatible Growth · May 2025



H.7 First SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
Asian Institute of Technology · 28 April–2 May 2025



H.8 Second SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
Asian Institute of Technology · 28 July–1 August 2025



SIPA
Sustainable Infrastructure Programme in Asia

Certificate of Participation

It is to certify that

Mr. Bernie C. Magtalas

Department of Economy, Planning and Development (DEPDev), Philippines
has participated in

The THIRD SIPA SOUTHEAST ASIA TRAINING SEMINAR ON SUSTAINABLE INFRASTRUCTURE DEVELOPMENT
held from 22nd to 26th February 2026 conducted by Centre for Global Challenges, Asian Institute of Technology, Thailand

Professor Deepak Sharma
Director, Centre for Global Challenges
Asian Institute of Technology
26 February 2026

UNDP OECD IKI

H.9 Third SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
Asian Institute of Technology · 22–26 February 2026



Unibersidad ng Pilipinas

Mabuhay!

Alinsunod sa batas ng Republika ng Pilipinas at batay sa rekomendasyon ng Kolehiyo at pagpapatibay ng Konseho ng Unibersidad, ipinagkaloob ng Lupon ng mga Rehente kay

Bernie Calderon Magtalas

na nakatupad sa mga itinakdang pangangailangan ng Kurso, ang titulong

Bachelor of Science in Electrical Engineering

kasama ang lahat ng mga karapatan, karangalan at pribilehiyo, gayundin ang mga obligasyon at pananagutang kaakibat nito. Ipinagkaloob sa Los Baños, Laguna, Pilipinas, ngayong ika-01 ng Agosto ng taong dalawang libo't labing-apat.

ALFREDO E. PASCUAL
Pangulo ng Unibersidad

LILIAN A. DE LAS LLAGAS
Kalihim ng Unibersidad

REX VICTOR O. CRUZ
Tanselor ng UP Los Baños

ARNOLD R. ELEPAÑO
Dekano ng Kolehiyo ng Inhenyeriya at Teknolohiyang Pang Agro-Industriyal

H.10 Diploma, Bachelor of Science in Electrical Engineering
University of the Philippines Los Baños · 2014

Submitted to : MAGTALAS Bernie Calderon
(Attention) This certificate is issued as a PDF with an electronic signature, and is not valid in printed form or if the name of the institution receiving the certificate cannot be identified.

No. 240F1658

WASEDA UNIVERSITY
TOKYO, JAPAN

(Student Number: 4022R349)

TRANSCRIPT OF ACADEMIC RECORD and CERTIFICATE OF COMPLETION

Graduate School: Graduate School of Asia-Pacific Studies
Major: Master MA Program in International Relations
Name: MAGTALAS, Bernie Calderon | Date of Birth:
Date of Admission: From September 21, 2022 | Date of Completion: September 15, 2024
Degree: Master of Arts in International Relations

SUBJECT	YEAR	Cr	Gr
[Core Courses]			
(Core Courses)			
Writing and Research Skills	2022	2	A
Analysis of Quantitative Data	2022	2	A+
Experiences in Global Cooperation of Japan	2022	2	A
Analysis of Quantitative Data II	2023	2	A+
Qualitative Research Methods	2023	2	A
Experiences in Development of Japan in International Society	2023	2	A
Multivariate Statistics	2023	2	A+
[Basic Courses]			
(Basic Courses)			
Environmental Economics and Environmental Policy	2022	2	A+
[Advanced Courses]			
(Advanced Courses (Group 2): International Relations)			
Energy Issues in Asia-Pacific	2023	2	A
(Advanced Courses (Group 3): International Development/ Policy Studies)			
Sustainable Development and International Development Cooperation	2022	2	A+
(Advanced Courses (Group 4): Japanese Studies)			
Japanese Economy	2023	2	A+
[Project Research: Master's Degree Program]			
(Project Research: MA Program)			
Economic Development and Policy in Asia (MA Seminar)	2024	8	A
[Courses Offered at Other Institutions]			
(Courses Offered at Other Waseda Graduate Schools)			
DATA SCIENCE FOR MANAGEMENT	2023	2	A+
[Non-credit Courses]			
(Non-credit Courses offered at Other Institutions)			
Japanese for 'Zero' Beginners	2022	2	B
***** End of Transcript *****			

SUBJECT	YEAR	Cr	Gr

Note: Grading System A+(100-90).A(89-80).B(79-70).C(69-60).P(100-60 or Completed)
We do not rank students in this school.
Subjects marked with * indicate credits transferred from other universities including overseas academic institutions.

TOTAL CREDITS 34

Cr:Credit Gr:Grade

NOTE

Date : October 14, 2024
President, Waseda University
TANAKA, Aiji

Signature

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H.11 Transcript of records, Master of Arts in International Relations
Waseda University · 2024



Capacitating youth leaders in bridging the achievement of the Sustainable Development Goals to the grassroots

PHILIPPINES

24 July 2017

MR. BERNIE C. MAGTALAS
Economic Development Specialist
National Economic and Development Authority
Ortigas Center, Pasig City

Dear **Mr. Magtalas**:

Greetings of solidarity from the 2030 Youth Force in the Philippines!

The 2030 Youth Force in the Philippines (2030 YFPH), is a youth-led organization anchored to the 2030 Youth Force in the Asia-Pacific Region and was organized by the United Nations Development Programme and United Nations Volunteers. Conscious of the youth's role in rebuilding societies, the 2030 YFPH is devoted to advance a principled national movement optimizing the potentials of the Filipino youth to contribute to the force that fuels and catalyzes the achievement of The Global Goals for Sustainable Development (SDGs) at the grassroots and marginalized sectors. In the long run, we envision an empowered Filipino Youth Sector working collectively towards the achievement of these goals for improved quality of life for all.

The 2030 YFPH governed by a Council, has 17 Global Goal Ambassadors, 17 Regional Representatives and over 300 member-advocates. Over the course of its operations, we saw the need to usher in passionate leaders who can reinforce the formation in undertaking its future engagements, projects, and campaigns. These young leaders are seen to be instrumental in expediting our operations as they will be working closely with the officers and be assigned of specific duties correspondingly.

With this, we are pleased to confirm your appointment as our new Co-Deputy Head for the Education and Research Committee. As the holder of the said position, you are expected to serve as one of the primary support officers of the Head for Education and Research. Your full support on the mandated function and responsibilities of your committee head is also expected, to wit (Art. IV, Sec. 6):

a. *Think-tank of the Youth Force;*
b. *Produce research and information materials and facilitate focus group discussions for the benefit of the members of the Youth Force;*
c. *Assist in resolving legal concerns;*
d. *Create contents of Social Media accounts of the Youth Force; and,*
e. *Perform duties as the Council deems necessary.*

Your respective committee head will reach out to you to get you further acquainted on the tasks that will be delegated to you. Meanwhile, our Head for Internal Affairs will facilitate your inclusion in all our online correspondences. You are highly encouraged to browse all previous communications and files.

We are looking forward to be working with you.

Thank you.

For the Global Goals,

Julliano Fernando A. Guiang
Convener
2030 Youth Force in the Philippines

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H.12 Appointment letter, Co-Deputy Head for Education and Research
2030 Youth Force in the Philippines Inc.

Acronyms and abbreviations

ADB
Asian Development Bank

AFD
Agence Française de Développement

AIIB
Asian Infrastructure Investment Bank

AIT
Asian Institute of Technology

AO
Administrative Order

APHC
Agus-Pulangi Hydropower Plant Complex

ASEAN
Association of Southeast Asian Nations

ASEP
Access to Sustainable Energy Programme

BAC-GIP
Bids and Awards Committee for Goods and Infrastructure Projects

BOO
build-own-operate

BOT
build-operate-transfer

BTOM
build-transfer-operate and maintain

CAAP
Civil Aviation Authority of the Philippines

CALAx
Cavite-Laguna Expressway

CBIT
Capacity-building Initiative for Transparency

CCAP
Climate Change Action Program

CFL
compact fluorescent lamp

CFPSS
Certified Flexible Power System Specialist

COP
Community of Practice

COVID-19
coronavirus disease 2019

CP3P
Certified Public-Private Partnership Professional

CTBEx
Cavite-Tagaytay-Batangas Expressway

DBCC
Development Budget Coordination Committee

DC
Department Circular

DED
detailed engineering design

DEPDev
Department of Economy, Planning, and Development

DHSUD
Department of Human Settlements and Urban Development

DOE
Department of Energy

DOTr
Department of Transportation

DPP
diesel power plant

DPPs
diesel power plants

EC
electric cooperative

EEC
energy efficiency and conservation

EICC
Energy Investment Coordinating Council

EIRR
economic internal rate of return

ENPV
economic net present value

EO
Executive Order

EPIRA
Electric Power Industry Reform Act of 2001

EPNS
Energy Projects of National Significance

ERC
Energy Regulatory Commission

EU
European Union

EV
electric vehicle

EVIDA
Electric Vehicle Industry Development Act

F/S
feasibility study

FEL-100
Future Energy Leaders 100 (World Energy Council)

FIRR
financial internal rate of return

FPC
fast patrol craft

GAD
Gender and Development

GCF
Green Climate Fund

GDP
gross domestic product

GEF
Global Environment Facility

GEMP
Government Energy Management Program

GGGI
Global Green Growth Institute

GIS
geographic information system

GOP
Government of the Philippines

HPP
hydroelectric power plant

IAEA
International Atomic Energy Agency

IAEECC
Inter-Agency Energy Efficiency and Conservation Committee

IBRD
International Bank for Reconstruction and Development (World Bank)

ICC
Investment Coordination Committee

ICPC
Iloilo Commercial Port Complex

IECC
Inter-Agency Energy Contingency Committee

IECC-OCTF
Inter-Agency Energy Contingency Committee – Oil Contingency Task Force

IED
Innovation Energie Développement

IIA
Iloilo International Airport

ILS
Integrated Logistics Support

InfraCom
Infrastructure Development Committee

IPP
independent power producer

ITS
intelligent transport system

JICA
Japan International Cooperation Agency

KEXIM
Export-Import Bank of Korea

LANDBANK
Land Bank of the Philippines

LED
light-emitting diode

LEDAC
Legislative-Executive Development Advisory Council

MCC
Millennium Challenge Corporation

MMITS
Metro Manila Intelligent Transport System

MPFRB
Multi-Purpose Fast Response Boat

MVIP
Mindanao-Visayas Interconnection Project

MWp
megawatt-peak

NBP
National Broadband Plan

NCR
National Capital Region

NEA
National Electrification Administration

NEDA
National Economic and Development Authority (now DEPDev)

NEECP
National Energy Efficiency and Conservation Program

NEP-IAC
Nuclear Energy Program Inter-Agency Committee

NEPIO
Nuclear Energy Program Implementing Organization

NIR
Negros Island Region

NLET
North Luzon Express Terminal

NOCECO
Negros Occidental Electric Cooperative

NONECO
Northern Negros Electric Cooperative

NORECO I
Negros Oriental I Electric Cooperative

NPC
National Power Corporation

O&M
operation and maintenance

OAT
operate-add-transfer

ODA
official development assistance

OECD
Organisation for Economic Co-operation and Development

OMUT
operate-maintain-upgrade-and-transfer

PAP
programs, activities, and projects

PAPs
programs, activities, and projects

PCG
Philippine Coast Guard

PGRDF
Philippine Geothermal Resource Derisking Facility

PhilATOM
Philippine Atomic Energy Regulatory Authority

PHP
Philippine peso

PLLO
Presidential Legislative Liaison Office

PNR
Philippine National Railways

POD
Port of Dumangas

PPIA
Puerto Princesa International Airport

PPP
public-private partnership

PRC
Professional Regulation Commission

PSALM
Power Sector Assets and Liabilities Management Corporation

PTB
passenger terminal building

PV
photovoltaic

PVM
Photovoltaic Mainstreaming

RA
Republic Act

RE
renewable energy

RENAC
Renewables Academy

RFID
radio-frequency identification

RIDI
Regional Infrastructure Development Index

RORO
roll-on/roll-off

SC-SDGs
Sub-Committee on the Sustainable Development Goals

SDG
Sustainable Development Goal

SDGs
Sustainable Development Goals

SHS
solar home system

SIPA
Sustainable Infrastructure Programme in Asia

SO
Special Order

SOP
Series of Projects

SPUG
Small Power Utilities Group

TA
technical assistance

TASF
Technical Assistance Special Fund

TEFI
Team Energy Foundation, Inc.

TEU
twenty-foot equivalent unit

TRAIN
Tax Reform for Acceleration and Inclusion

TWG
technical working group

UK
United Kingdom

UNDP
United Nations Development Programme

US
United States

USD
United States dollar

WTE
waste-to-energy